

AYUSH GUPTA

Bhubaneswar, Odisha, India

PROFESSIONAL SUMMARY

Third-year B.Tech Computer Science Scholar at KIIT University with hands-on experience in full-stack web development and machine learning projects. Skilled in building scalable backend systems and interactive frontend applications using modern web technologies. Enthusiastic about problem-solving, and applying AI/ML techniques in real-world projects. Actively involved in coding practice, project development, and continuous learning in software engineering.

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python

Frontend: React.js, HTML, CSS, JavaScript

Backend: Node.js, Express.js, REST API Development, WebSocket

Databases: MongoDB, SQL, ChromaDB

DevOps & Version Control: Docker, Kubernetes, Git, GitHub

AI / Machine Learning: NumPy, Pandas, Scikit-learn, Flask, Supervised Learning, Unsupervised Learning, Reinforcement Learning, Deep Learning, Agentic AI, LLMs, GANs, RAG

Data Visualization: Power BI, Matplotlib, Seaborn

EDUCATION

Bachelor of Technology – Computer Science Engineering | **KIIT University** | Bhubaneswar, Odisha

- CGPA: 9.77 / 10.00 | Scholar Student (merit-based scholarship) *Aug 2024 – 2028*

Senior Secondary (Class XII) | **Don Bosco School** | Guwahati, Assam

- Percentage: 85% *Aug 2021 – 2023*

Secondary (Class X) | **Don Bosco School** | Lumding, Assam

- Percentage: 89.5% *Jan 2009 – 2021*

PROJECTS

AI/ML + Power BI Crime Analytics Dashboard | **Personal Project** | Deployed

- Built an interactive React.js dashboard integrating Power BI visuals with AI/ML predictive models on the India Crimes Dataset (NCRB source) comprising 13,066 records
- Applied supervised and unsupervised ML algorithms (Scikit-learn) for crime trend analysis, pattern detection, and state-level forecasting
- Deployed full-stack application with REST API backend (Flask) serving model inference results to the frontend
- Technologies: React.js, Python, Flask, Scikit-learn, Power BI, Pandas, Matplotlib

Airbnb-Like Rental Platform Clone | [Personal Project](#) | 2026

- Developed a full-featured property listing and booking platform using MERN stack, replicating core Airbnb functionalities
- Implemented RESTful APIs, JWT authentication, image upload, and search/filter functionality
- Technologies: MongoDB, Express.js, React.js, Node.js, REST API

Physics-Based Simulations – Unreal Engine 5 | [Personal Project](#) | 2025

- Built 3 physics-based interactive simulations including particle and VFX work using Niagara system in UE5
- Implemented realistic collision, fluid dynamics, and environmental effects for visual demonstrations

Smart Agriculture IoT Simulation (Research) | [Independent Research](#) | 2025

- Developed a Python-based soil moisture diffusion and evaporation simulation modelling real-world irrigation physics
- Simulated IoT sensor noise and implemented a proportional controller for adaptive irrigation regulation
- Visualized moisture depth profiles over time using Matplotlib; explored integration with ESP32 microcontroller concepts
- Technologies: Python, NumPy, Matplotlib, Physics-based Modelling, IoT (ESP32 concepts)

Mini Web Applications Suite | [Personal Projects](#) | Deployed | 2025

- Developed multiple standalone web apps: To-Do List, Weather App (OpenWeatherMap API), and browser-based mini-games
- Focused on clean UI/UX, API integration, and responsive design principles

HACKATHONS & COMPETITIONS

- Smart India Hackathon (SIH) 2025 – Participated as part of a cross-functional team; developed solution under national-level innovation challenge
- IIT Guwahati Techniche 2022 – Participated in technical events at one of India's premier technology fests
- Multiple hackathons across AI/ML and web development domains; built end-to-end prototypes under time-constrained environments

ADDITIONAL INFORMATION

Languages: English , Hindi , Assamese , Bengali , Bhojpuri

Interests: AI Workflow Automation, MCP, LLM APIs, Competitive Programming

Contact: +91 9678954733

Email: ashzex0@gmail.com / 24052068@kiit.ac.in

GitHub: github.com/theashverse

LinkedIn: linkedin.com/in/ayush-gupta-58819527a